

Validation of an AI Model for Cerebral Aneurysm Detection in MR Angiography : Comparative Analysis of MR-aneurysm AI and Radiologists

Purpose

The utilization of artificial intelligence (AI) in medical imaging has advanced significantly, with AI tools increasingly adopted to assist in diagnostic processes. While substantial progress has been made in AI-based analysis of brain lesions using modalities such as CT and conventional MR imaging, limited efforts have focused on MR angiography (MRA). This is primarily due to MRA's susceptibility to artifacts and noise, as well as the complexity of accurately isolating cerebral vessels from other anatomical structures.

In this study, we developed an AI model designed to automatically extract cerebral arteries and detect cerebral aneurysms from MRA. To validate its performance, we compared the model's diagnostic accuracy against that of experienced radiologists using an external validation dataset. Advanced image processing techniques and machine learning algorithms were incorporated to enhance the robustness of the AI model under challenging conditions.

Materials and Methods

The training dataset consisted of MR-TOF (time-of-flight) sequence images collected from 3,500 aneurysm patients and 300 negative controls across two institutions, under IRB approval (P01-202305-01-042). After excluding unsuitable data, such as images with metal artifacts, MIP reconstructions, and wide slice intervals, the final dataset comprised 4,360 aneurysms from 3,036 MR-TOF images and 278 negative controls. These images served as the foundation for developing a robust AI model tailored for cerebral aneurysm detection.

Firstly, an advanced MR-vessel AI model was developed to extract cerebral arteries from MR-TOF (time-of-flight) sequence images and generate high-resolution 3D vascular maps with minimal artifact interference. Using the MR-vessel AI model, 3D vascular maps were reconstructed from 3036 MR-TOF images. An experienced neurosurgeon segmented aneurysms from these maps, resulting in 4,360 annotated aneurysms.

Secondly, the segmented MR-TOF images were divided into 3D sub-volumes and processed by the AI model, which predicted aneurysms in each sub-volume. The model's outputs were refined through post-processing techniques to combine predictions and accurately identify clinically relevant aneurysms. This iterative process culminated in the development of the MR-aneurysm AI model, fine-tuned to achieve optimal performance.

For external validation, 100 MR-TOF images containing 150 aneurysms confirmed by digital subtraction angiography (DSA) were independently analyzed by the MR-aneurysm AI model and three neuroradiologists with over five years of MRA interpretation experience. Detected aneurysms were categorized by location into 19 anatomical sites, including bilateral lesions (paraclinoid artery, ophthalmic A, PComA, AChOA, ICA bifurcation, MCA, PCA, VA) and unilateral lesions (ACoMA, DACA, basilar artery). Diagnostic performance was assessed using sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV).

Results

Each MR-TOF image was divided into 19 anatomical locations for analysis, resulting in a total of 1,900 aneurysm locations evaluated. These included 150 locations with confirmed aneurysms (lesion group) and 1,750 locations without aneurysms (control group).

Radiologist A detected 113 aneurysms, achieving a sensitivity of 66.0% and specificity of 99.2%. Radiologist B detected 133 aneurysms, with a sensitivity of 72.7% and specificity of 98.6%. Radiologist C identified 111 aneurysms, yielding a sensitivity of 67.3% and specificity of 99.4%. When combined, the radiologists exhibited a collective sensitivity of 68.7%, specificity of 99.1%, positive predictive value (PPV) of 86.6%, and negative predictive value (NPV) of 97.4%.

In comparison, the MR-aneurysm AI model identified 177 suspected aneurysms, of which 138 were confirmed as true positives. This corresponded to a sensitivity of 92.0%, specificity of 97.8%, PPV of 78.0%, and NPV of 99.3%. Statistical analysis demonstrated that the AI model had significantly higher sensitivity than the radiologists (p-value: 0.0324).

A subgroup analysis further highlighted the superior sensitivity of the AI model across the five most common aneurysm locations: paraclinoid artery, anterior communicating artery (ACoMA), middle cerebral artery (MCA), posterior communicating artery (PComA),

and anterior choroidal artery (AChoA). For instance, among 50 paraclinoid aneurysms confirmed by DSA, the AI model detected 48 (96.0%) compared to 32.7 (65.3%) by radiologists. Similarly, for AComA aneurysms, the AI model achieved a sensitivity of 87.5%, outperforming the radiologists' sensitivity of 79.2%. The differences in sensitivity between the AI model and radiologists were statistically significant (p-value: 0.0062).

Conclusion

The MR-aneurysm AI model demonstrated superior diagnostic performance compared to radiologists, particularly in terms of sensitivity (92.0% vs. 68.7%; p-value: 0.0324). Furthermore, the AI model exhibited a better ability to detect cerebral aneurysms compared to radiologists, especially in cases involving high prevalence. Notably, the AI model's enhanced sensitivity highlights its potential clinical utility, despite its relatively lower PPV (78.0% vs. 86.6%).

However, limitations such as the small size and biased nature of the validation dataset warrant further study. Expanding the dataset to include a greater diversity of aneurysm types and locations, particularly vertebral artery and PCA aneurysms, is expected to further enhance the model's performance and applicability in clinical practice.